

Public vessel-tracking data does not predict the Henry Hub–TTF spread: a pre-registered, power-bounded null

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Abstract

Positions of liquefied natural gas carriers are widely believed to carry information about the price gap between North American and European gas. A cargo that leaves the US Gulf today lands in Europe in two to three weeks, and that supply does not yet appear in any published statistic. We test this belief. From 54,494,686 AIS position fixes (2016-01-01 to 2026-08-10) we reconstruct 87,874 port events and a 34-signal daily panel, built once with hindsight for validation and once strictly point-in-time for modelling. We then ask whether tanker-derived signals predict the one- and four-week change in the Henry Hub–TTF spread after controlling for its own persistence, weather, storage and oil. Each hypothesis, sign and acceptance bar was recorded in a dated append-only log before the corresponding fit. We find no predictive content. A controls-only model performs worse than a no-change forecast (−9.3% at one week, −18.2% at four). No signal reaches a Newey–West $|t|$ of 1.96 (maximum 1.50). Three pre-registered mechanisms that a linear scan could miss (the Europe-bound share of cargo, an interaction with EU storage, and a tightness-regime split) fail under a Bonferroni bar with an untouched holdout. The design can detect effects of at least 1.72% of residual variance; the largest observed is 0.009. The result therefore excludes effects large enough to trade and says nothing about smaller ones. We also document two construction hazards specific to voyage-derived features. The unit of observation is defined only in retrospect, and feature availability changes across data vintages. A naive loader leaked 65% of the sample.

1 Introduction

The Henry Hub–TTF spread is the price of moving a unit of gas from the United States to Europe, and liquefied natural gas (LNG) carriers are the means by which it moves. A laden carrier that leaves Sabine Pass arrives at Rotterdam roughly fourteen to eighteen days later. Between departure and arrival its cargo is committed, is visible to anyone with an AIS receiver, and is absent from every published supply statistic. It is therefore natural to expect vessel positions to lead the spread. This expectation is widely held and is the basis of a commercial data industry.

This paper tests it and finds no support. An efficient-markets view predicts that outcome, so the null itself is not the contribution. The contribution lies in how the null is established, and in two construction problems encountered on the way.

Three features of the design distinguish the result from a mere absence of findings. First, every target, control set, lag, sign and acceptance bar was written into a dated, append-only decision log before the corresponding fit, so no choice reported below was made after seeing its outcome. Where a pre-registered sign was falsified, it is reported as falsified. Second, the

inference respects the structure of the data. Weekly changes over a four-week horizon overlap, so all standard errors are Newey–West with the lag fixed in advance at $h + 1$ weeks. Training rows whose target reaches into the test week are purged. The three follow-up mechanism tests are held to a Bonferroni bar and must replicate their sign on a holdout that was not examined until they had passed on the discovery sample. Third, the study bounds what it could have found. With 327 weekly observations the design detects effects of at least 1.72% of residual variance, and the largest observed anywhere is 0.009. The result excludes effects large enough to trade and is silent on smaller ones.

Two further findings are methodological and apply to any feature built from vessel-tracking data. The natural unit of observation, a voyage, does not exist until both of its endpoints have been observed, so a statistic computed over completed voyages embeds the future unless the population is explicitly bounded. At a 2020 as-of date a straightforward loader returned 20,351 legs, of which 13,130 (65%) had been closed by arrivals that had not yet occurred. Separately, the availability of a feature can change across data vintages. The same loader attached 2,101 destination declarations to 2020 voyages from a data feed that began in 2026. A date filter on observations cannot catch this, because the individual rows are correctly dated; the table itself should not exist at that vintage.

The paper proceeds as follows. Section 2 describes the pipeline and the decade-scale reconstruction. Section 3 explains how the signal panel is built point-in-time. Section 4 describes the pre-registration protocol and its limits. Section 5 reports the physical nowcasts. Section 6 reports the spread tests. Section 7 bounds their power. Sections 8 and 9 interpret and qualify the results.

2 Data and pipeline

2.1 Sources

Vessel positions come from three feeds of different fidelity, and no signal is aggregated across the boundary between them. For the United States we use the NOAA Marine Cadastre archive of terrestrial Class-A AIS [10], which is exhaustive along the US coast from 2016 and ends on 2025-12-31. For Europe and the open ocean we use Global Fishing Watch port-visit and voyage events [5]. These supply arrivals and departures but carry no coordinates and no anchorage events; the feed ends on 2026-02-21. A live subscription to an AIS aggregator covered a tier-ranked fleet from April 2026 and is used only for the vintage self-validation described in Section 3.

Prices and controls are public. Henry Hub spot and Lower-48 storage are from the EIA API [12]. The TTF front-month contract is from a keyless quote feed and was cross-checked against the World Bank’s monthly European gas series [13]; the regression slope is 0.999 with $R^2 = 0.9999$ over 106 months, so the distortion introduced by the continuous roll is negligible. EUR/USD and Brent are from FRED [2]. EU storage is from GIE AGSI+ [4]. Heating and cooling degree days are computed from ERA5 reanalysis temperatures at weighted demand centres [11].

2.2 Reconstruction

Table 1 gives the scale of the reconstruction. Fixes are resolved against berth, anchorage and approach polygons at every LNG terminal in seven zones (US Gulf, US Atlantic, north-west Europe, Baltic, Iberia, western Mediterranean and eastern Mediterranean) by a per-vessel state machine that emits zone entry, anchorage, mooring and departure events. Three pairings of

Table 1: Reconstruction inventory. Port events are labelled by the feed that produced them. The two backfill feeds end months before the live feed, which determines the sample choice in Section 6.

layer	rows	span
AIS position fixes	54,494,686	2016-01-01 to 2026-08-10
port events (state machine)	87,874	
noaa	44,363	to 2025-12-31
gfw	40,273	to 2026-02-21
mmsi_filter	1,974	to 2026-08-10
bbox	1,264	to 2026-05-30
in-scope fleet (LNG carriers + FSRUs)	828	
laden US export departures	9,855	
signal panel rows (<code>signal_daily</code>)	1,032,982	34 keys, two bases
model panel rows (<code>model_panel</code>)	77,500	20 features

Table 2: NOAA-only capture of US LNG export cargoes relative to the EIA-implied count. The project’s diagnostic tool counts every feed and therefore double-counts departures seen by both NOAA and GFW; the series reported here uses the single feed.

year	captured	EIA-implied	capture
2016	48	51	95%
2017	159	192	83%
2018	216	294	74%
2019	290	494	59%
2020	300	648	46%
2021	715	966	74%
2022	918	1048	88%
2023	1,099	1178	93%
2024	1,216	1185	103%
2025	1,570	1494	105%

those events underlie everything that follows. A *leg* pairs a departure with the vessel’s next zone entry and represents a voyage. A *visit* pairs a mooring with the next departure and represents berth occupancy. A *queue* pairs the first anchorage entry with the next mooring at the same terminal and represents the wait to berth. Laden state is inferred from reported draught relative to design draught.

2.3 Ground truth

The US side of the reconstruction can be checked against a published total. The EIA reports monthly US LNG exports. Dividing by a nominal cargo of 174,000 m³ gives an implied cargo count, and the ratio of observed laden departures to that count is a capture rate (Table 2). Restricted to the NOAA feed, the rate is 46% in 2020 and rises to 93%, 103% and 105% in 2023, 2024 and 2025. The gradient reflects the coverage of the archive in its early years. Early-year levels of any US signal are therefore biased low by up to half. The models below work on within-window changes or on standardised values, and none treats an early-year level as a physical quantity.

Figure 1 shows the target and the headline signal on one timeline. The two structural events that any model must survive are visible: the 2020 and 2021 inversions, in which Henry Hub briefly exceeded TTF, and the 2022 European crisis.

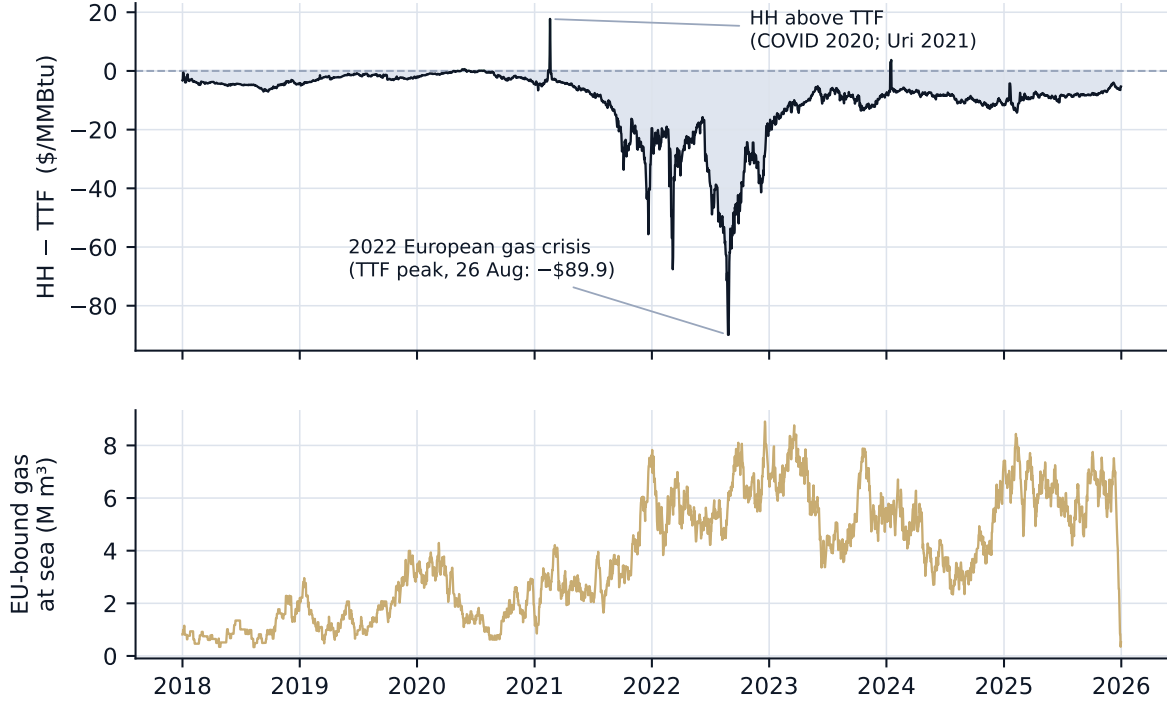


Figure 1: The Henry Hub-TTF spread (top) and EU-bound gas at sea (bottom), daily, on the coverage-consistent sample used for modelling. The extremes coincide with Winter Storm Uri (February 2021) and the August 2022 TTF peak.

3 Building the panel point-in-time

Every signal is built twice from the same events. A *physical* basis uses hindsight and serves validation. A *knowable* basis uses, on each day d , only what could have been observed on day d , and serves modelling. Two properties of voyage-derived features make the knowable basis difficult to construct correctly. Both were discovered during the project rather than anticipated.

3.1 The unit of observation is defined in retrospect

A voyage leg does not exist until both a departure and the next arrival have been observed. At time t , a vessel that departed five days earlier has an incomplete leg of unknown duration and unknown destination. In hindsight the same vessel has a complete leg with both. Any statistic over legs therefore conditions on completion unless the population is explicitly bounded. Figure 2 quantifies the effect. At an as-of date of 1 June 2020, a loader that pairs all departures with all arrivals and then filters by date returns 20,351 legs. Of these, 13,130 (65%) are closed by arrivals that had not yet occurred. With the population bounded correctly, 4,713 legs remain, none of them closed by the future. Status labels move with the bound: the number of legs classified as floating storage fell from 95 to 9, because that label is assigned in hindsight. At an as-of date after the data maximum the two loaders agree on every count, which confirms that the bound is a restriction of the same query and not a different one.

3.2 Feature availability changes across vintages

The same naive loader attached 2,101 destination declarations to the 2020 legs. Declarations come from a live AIS static-data feed that began in 2026, so no declaration existed in 2020.

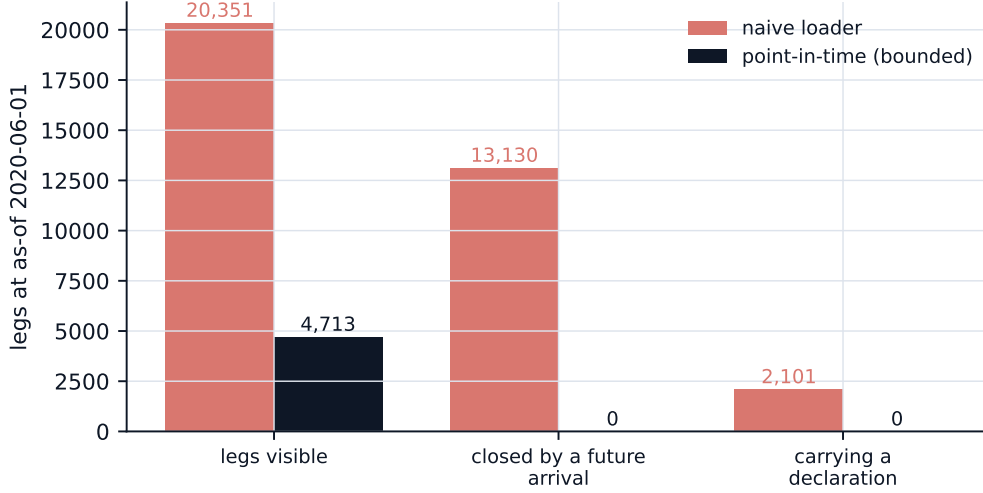


Figure 2: Voyage legs visible at an as-of date of 1 June 2020 under the naive loader and under the point-in-time bound. Recomputed from the event tables when the figure is generated.

This failure differs from the first. A date filter on the observations is satisfied, because each declaration row carries a timestamp within the vessel’s current voyage, but the table being joined should not exist at that vintage. The bound therefore reconstructs the declaration from the raw static messages available at d , and at a 2020 as-of date it returns none.

3.3 What the knowable basis bounds

The knowable basis bounds on event time. Event timestamps are back-dated to the fix at which a transition first qualified, so a departure is dated to the vessel’s last fix inside the berth polygon and not to the later fix that confirmed it had left. The gap between the two is the feature’s own publication lag. We measured it directly. The first fix after a departure event arrives at a median lag of zero hours, and more than 24 hours later for 0.3% of departures on the NOAA feed ($n = 1,500$) and 1.5% on the live feed ($n = 343$). On the decade panel the lag is immaterial.

A separate log records the knowable value of every signal as it was printed on each day of live operation, so that the bound can be checked rather than assumed. Comparing the rebuilt knowable series with those prints on days old enough to have settled, the check flags 68 rows whose value moved after a next-day print. All of them lie in the live tail and are consistent with late-arriving fixes back-dating departures. A further 909 rows differ from prints taken during the day, before that day’s fixes had finished arriving. That comparison measures intraday timing and not leakage; it is reported but not scored. The live tail lies outside the modelling sample used in Section 6.

3.4 The modelling grid

Prices, storage and the tanker signals arrive on different calendars. They are assembled onto one daily grid by forward-fill under a per-cadence staleness ceiling. An inner join is never used: Henry Hub spot prints on 69% of calendar days, and an inner join across the price legs would drop a third of the history without any indication that it had done so. Each external series is shifted to the date on which it became public (EIA weekly storage by six days, daily spot by one, AGSI+ by one). The tanker signals are shifted by zero, because the knowable basis is already point-in-time and a second shift would double-count. Degree days are not shifted.

Table 3: Pre-registration entries. The commit column is discussed in the text.

entry	dated	what was fixed before the fit	commit
D-001–D-003	2026-08-10	A1 target, kernel, replay protocol	4239d13
D-016	2026-08-12	A2 design lock	4239d13
D-019	2026-08-12	A5 design lock	4239d13
D-022	2026-08-12	A4 design lock	4239d13
D-025	2026-08-12	H1 horizon test pre-specified	4239d13
D-028	2026-09-05	Part B null + FWL: target, controls, HAC lag, signs	e483a1b
D-031	2026-09-05	H2/H3/H4: signs, Bonferroni bar, holdout	c489550

Their role is to be partialled out, so the relevant quantity is the weather that occurred and not the weather that was forecast. Signals reported per terminal or per destination zone are collapsed to one series by summing volumes and by n -weighted averaging of durations.

3.5 The coverage seam

The two backfill feeds end on 2025-12-31 (NOAA) and 2026-02-21 (GFW). After those dates the signals are carried by the live feed alone, which contributes 3,238 events against 84,636 from the backfills. The EU-bound stock at sea falls from 4.80 to 0.41 million m³ between December 2025 and January 2026, a 91% drop with no physical cause. This was found by plotting Figure 1 and was not caught by any automated check. The primary modelling sample is therefore truncated at 2025-12-31. Full-panel results are reported in Appendix B and change no conclusion.

4 Pre-registration protocol

The project keeps a dated, append-only decision log. A hypothesis, target, lag, sign or acceptance bar is recorded on the day its entry is written and before the corresponding fit. A later change is made by a new, dated entry that names the result which prompted it. The log is what allows a reader to distinguish a test that found nothing from a search that stopped at the first result to cross a threshold. Table 3 lists the entries that fixed each model reported below.

Two consequences of the protocol appear in the results. A pre-registered sign that the data falsified is reported as falsified (Section 5). A test that produced an apparent 87% improvement over its null was withdrawn, because the protocol’s own comparability condition failed (Section 5). Neither outcome would have been recorded without a criterion fixed in advance.

Limits of the log. The log is a text file in the project’s version control, and its git history does not independently establish the ordering it asserts. The entries for the physical nowcasts (D-001 through D-026) were committed together, weeks after the work they describe. The two spread pre-registrations (D-028 and D-031) were each committed in the same change as their result. The content of each pre-registration, meaning the signs, bars, horizon and holdout fixed in its text, was written before the corresponding run, but a sceptical reader has only the author’s word for this. We state it here rather than leave it to be found. A third-party timestamped registry would have cost nothing and will be used for any future entry, which will be committed and pushed before the run it governs.

Table 4: One-week-ahead nowcasts of weekly US loadings (A2, A4) and conditional EU arrivals (A1), scored walk-forward from 2018 on the NOAA feed. MAE is in cargoes per week. All three fail the pre-registered bar of beating both nulls.

model	MAE	persistence	climatology	vs climatology	80% cov.
A1 arrival-count climatology	2.314	2.217	1.814	+27.5%	73.7%
A2 negative-binomial GLM	3.961	2.551	2.152	+84.0%	75.7%
A4 Kalman local level	2.104	2.694	2.088	+0.7%	72.6%

Table 5: Outage detection against labelled terminal outages. N2 is the pre-registered null and the usable artefact; its recall is the relevant number.

detector	recall	median delay (d)	false alarms / terminal-yr
BOCPD (pre-registered)	12% (2/16)	17	0.13
N1 absolute silence (14 d)	88% (14/16)	14	2.33
N2 rate-relative silence	38% (6/16)	12	0.19

5 Physical nowcasts

Before asking whether the signals predict a price, we ask whether they predict the physical quantities they measure directly: weekly EU arrivals, weekly US loadings and terminal outages, at one- and two-week horizons. These targets have high signal-to-noise ratios and published ground truth, and they establish what the pipeline can and cannot see before any price enters the analysis. Four models were pre-registered and scored walk-forward against two naive nulls, persistence and a four-week climatology. Each model was designed to address the diagnosed weakness of the previous one. Table 4 gives the one-week results.

The sequence of results is informative. A1, a duration-climatology convolution of the at-sea stock, is +27.5% worse than the climatology null; its estimator is maturity-gated and therefore stale. A2, a negative-binomial count model on lagged loadings and berth occupancy, is +84.0% worse because it over-extrapolates the trend. A2 also falsified one of its pre-registered signs: ballast arrivals entered with coefficient -0.049 against a prior of positive. A4, a Kalman local-level model, is +0.7% from the null, which we treat as a tie. The tie carries information. A4 fitted its smoothing constant by maximum likelihood and chose $\alpha = 0.251$, an effective window of 7.0 weeks. For a series that is a local level plus noise, the exponentially weighted moving average is the optimal linear predictor [8]. The fitted model landing on a four-week simple average indicates that the optimum was found, and not that four methods happened to tie. On this target and at this horizon, the naive mean is close to optimal, and a fifth method would not have changed the conclusion.

Outage detection. A5 tested Bayesian online change-point detection [1] on the inter-departure process at each US terminal against 16 labelled outages over 54 terminal-years (Table 5). It did not outperform a rate-relative silence rule, which detects with a median delay of 12 days at 0.19 false alarms per terminal-year. The rule’s recall is 38% (6/16). It is a usable but limited monitor, and it is the baseline rather than the proposed model.

A withdrawn test. A natural extension asks whether skill rises with horizon, since cargo at sea today determines arrivals two to three weeks ahead. This was pre-specified before any longer-horizon data was examined. Extending A1’s window to four weeks produced a +86.7% improvement over the best null at week four (Table 6). The improvement is an artefact. A1’s target is conditional on legs already at sea, whereas both nulls are the last elapsed unconditional

Table 6: A1 extended to four weekly windows. The skill column at W3 and W4 is an artefact of a conditional target scored against unconditional nulls; the harness no longer reports it as a result.

window	mean truth	A1 MAE	persistence	climatology	“skill”	80% cov.
W1	5.89	2.324	2.226	1.822	−27.5%	73.5%
W2	5.49	2.121	2.052	1.773	−19.6%	76.0%
W3	1.98	1.170	4.146	4.058	+71.2%	85.9%
W4	0.75	0.691	5.247	5.208	+86.7%	94.2%

weekly count. The two coincide at one and two weeks, because no US-to-EU voyage completes in under seven days. Past the voyage tail the conditional truth falls toward zero (mean 5.89 arrivals in week one and 0.75 in week four) while the nulls continue to predict the full weekly rate (5.25 MAE). A1 was forecasting a smaller quantity, not forecasting better. The pre-specified premise, that the nulls degrade with horizon, fails, and the protocol treats that outcome as grounds to abandon the test. In the comparable region, skill moves from −27.5% to −19.6%, which is the predicted direction but remains below the null.

6 The spread

6.1 Target, grid and controls

The target is the spread $s_t = \text{HH}_t - \text{TTF}_t$ in \$/MMBtu, with TTF converted from EUR/MWh at the day’s EUR/USD rate and 3.412 MWh per MMBtu. It is sampled weekly on Mondays from 2018, the same grid as the physical nowcasts. The scored quantity is the h -week-ahead change, $\Delta_h = s_{t+7h} - s_t$, for $h \in \{1, 4\}$. The level is too persistent to score usefully, because a model that predicts the current value performs well on it. The primary sample runs to 2025-12-31 (Section 3) and contains 379 complete weeks.

The control block consists of the spread’s non-tanker drivers: heating and cooling degree days for the US and north-west Europe, US and EU storage, Brent, and a winter dummy. The price legs that construct the target are excluded as regressors. All standard errors are Newey–West [9] with the lag fixed in advance at $h + 1$ weeks. On a weekly grid with $h = 4$, consecutive targets share three weeks, and an ordinary least-squares standard error is roughly $\sqrt{h}$ too small. Fits are walk-forward with an expanding window and a 104-week minimum. A training row is admissible only if its target closed strictly before the test week, with a further one-week embargo [6].

6.2 The controls-only null is worse than no change

Three models form a ladder in which each is the null for the next. M0 predicts no change. M1 is an AR(1) in the level. M2 adds the controls and is the AR(1)+controls null that the tanker signals were pre-registered to beat. Table 7 shows that M2 does not beat M0. Relative to a no-change forecast, M2 scores −9.3% at one week and −18.2% at four. It is worse at the longer horizon and worse in each of 2023, 2024 and 2025 taken individually. This is the behaviour expected of an efficient and persistent price series: the level already incorporates the fundamentals, so adding them as regressors introduces estimation noise. The operative null for the signals is therefore M0, a stricter bar than the one pre-registered.

Table 7: The model ladder on the primary sample. Skill is the fractional MAE improvement over M0; a negative value means worse than a no-change forecast.

h (wk)	model	n	MAE	RMSE	80% cov.	skill vs M0
1	M0	272	2.034	4.465	81.6%	—
1	M1	272	2.061	4.571	81.6%	−1.4%
1	M2	272	2.223	4.630	82.4%	−9.3%
4	M0	266	3.942	7.203	72.9%	—
4	M1	266	4.196	7.585	70.7%	−6.4%
4	M2	266	4.660	7.437	69.5%	−18.2%

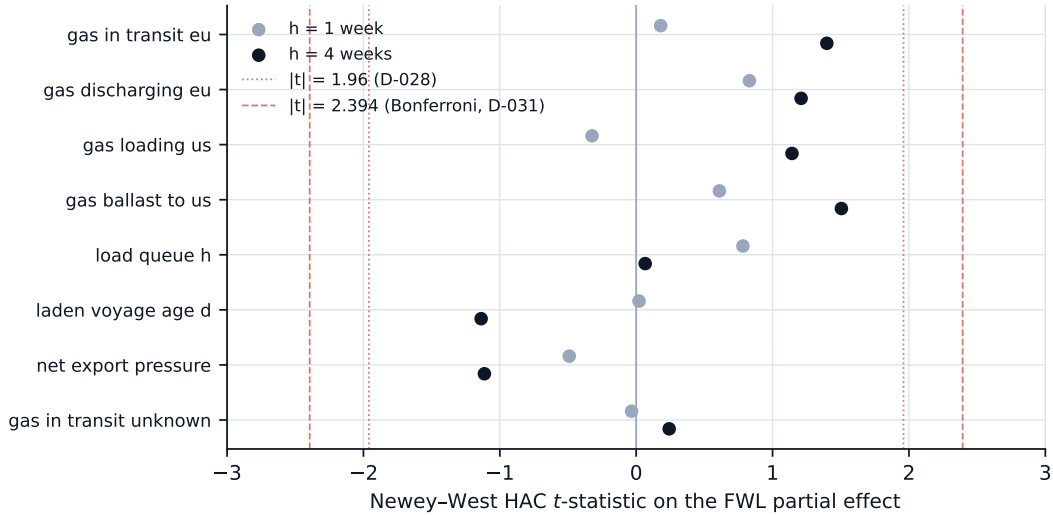


Figure 3: Newey–West t -statistics on the FWL partial effect of each pre-registered signal, after controlling for persistence, weather, storage and oil, at one and four weeks. The dotted and dashed lines mark the two decision bars used in the paper.

6.3 No signal survives partialling

For each of eight pre-registered signals we residualise both the target and the signal on M2’s regressors and regress residual on residual. This is the Frisch–Waugh–Lovell partial effect [3, 7]. By the theorem it equals the signal’s coefficient in the full regression, and that equality serves as the harness’s correctness test. Each signal carried a sign fixed in advance (Appendix A). A result counted only if $|t| > 1.96$, the sign matched, and the sign held in at least two-thirds of individual years. The last condition guards against 2022 alone driving a pooled fit.

No signal meets the bar (Figure 3, Table 8). The largest $|t|$ at either horizon is 1.50, and the largest partial R^2 is 0.009. The most direct mechanical hypothesis, that cargo at sea today lands in Europe in two to three weeks, was tested as the EU-bound stock and gave $t = +0.18$ at one week and $+1.40$ at four. At four weeks the four flow signals all have their pre-registered sign, but all lie well inside the noise band. They are directionally consistent with the mechanism and uniformly insignificant. No sign was falsified, because no estimate reached significance.

6.4 Three mechanisms a linear scan could miss

A one-at-a-time linear scan has three identifiable blind spots. Each was tested with a derived feature or a sample split, so each costs a single test. The three were pre-registered together with a Bonferroni bar of $|t| > 2.394$, a primary horizon of four weeks chosen from the voyage

Table 8: FWL partial effects on the primary sample. Year consistency is the share of individual years whose partial effect has the full-sample sign.

signal	prior	$h = 1$		$h = 4$		year cons.
		HAC t	partial R^2	HAC t	partial R^2	
gas in transit, EU-bound	+	+0.18	0.000	+1.40	0.009	62%
gas in transit, unresolved	none	-0.03	0.000	+0.24	0.000	50%
US loading rate	+	-0.32	0.000	+1.14	0.006	62%
EU discharge rate	+	+0.83	0.001	+1.21	0.003	62%
ballast returning to US	+	+0.61	0.001	+1.50	0.008	62%
laden voyage age	-	+0.02	0.000	-1.14	0.004	62%
US load-queue wait	-	+0.78	0.001	+0.07	0.000	38%
net export pressure	+	-0.49	0.000	-1.11	0.003	75%

Table 9: The three pre-registered mechanism tests on the discovery sample at $h = 4$ with Bonferroni bar $|t| > 2.394$. None is significant. H3 has the wrong sign, and H4’s tight-regime effect is smaller than its loose-regime effect.

hypothesis	prior	$\hat{\beta}$ (discovery)	HAC t	year cons.	partial R^2	$\hat{\beta}$ (holdout)
H2: EU share of at-sea stock	+	-2.64×10^0	-0.89	71%	0.003	-2.51×10^0
H3: transit $\times$ EU storage	-	4.78×10^{-9}	+0.46	71%	0.001	-1.49×10^{-9}
H4: transit, tight regime	+	-1.13×10^{-7}	-0.17	100%	0.000	2.71×10^{-7}
H4: transit, loose regime	+	-2.25×10^{-7}	-1.00	100%	0.009	-

mechanism, and a holdout (2025-01-06 to 2025-12-01, 48 weeks) that was not examined until a hypothesis had passed on the discovery sample (2018-01-01 to 2024-12-30, 327 weeks). A pass required the holdout sign to match.

H2 tests the share of cargo bound for Europe in place of the level. Most of the at-sea stock has no resolved destination, so two collinear levels with offsetting effects could each appear insignificant while their ratio carries the signal. A pre-registered gate voids the test if the share is mostly a decade-long trend, which would indicate improving coverage rather than changing cargo routing. The trend explains $R^2 = 0.209$ and the gate passes. H3 tests an interaction with EU storage. Additional cargo arriving when storage is 90% full has little effect, while the same cargo at 30% is highly price-relevant, and a linear fit averages the two states. H4 tests a tightness split, the simplest form of a regime model. The scan is re-run within weeks in which EU storage is below its seasonal norm (with the norm fitted on the discovery sample only), and the prediction is that the effect is positive in those weeks and larger than in the remaining weeks.

All three fail (Table 9), and none approaches the bar. The largest $|t|$ among the primary tests is 0.89. Each failure argues against a further attempt. The H2 estimate is stable and negative on both discovery (-2.64×10^0) and holdout (-2.51×10^0), which is the wrong sign, and it never approaches significance. H3 is positive ($t = +0.46$) against a prior of negative, so neither the magnitude nor the direction of the storage interaction is present. H4’s tight-regime effect is negative ($t = -0.17$) and smaller in magnitude than the loose-regime effect, which reverses both predictions. On this evidence, cargo arrivals do not have a measurable effect in either storage state.

The linear null of the previous subsection has thus been tested against the three specific ways in which it could have been wrong, using pre-registered signs, a corrected bar and an untouched holdout, and it stands.

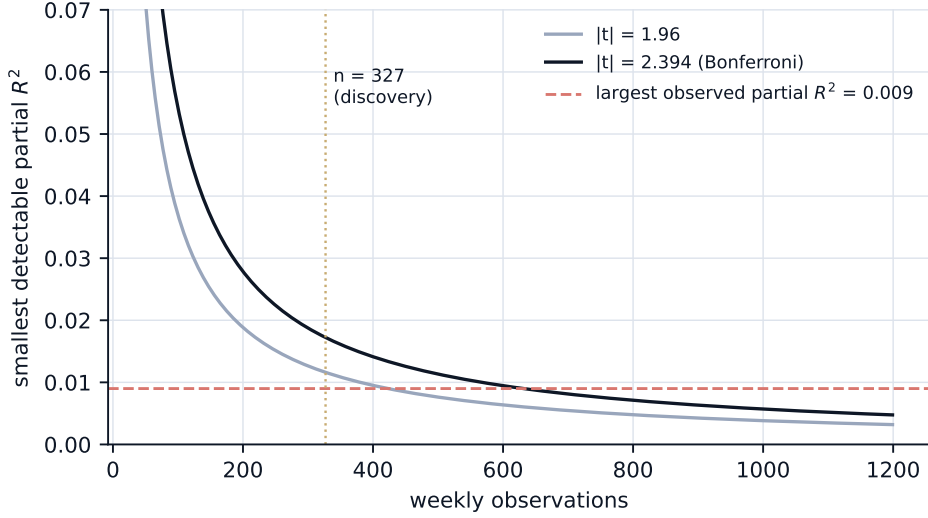


Figure 4: Smallest detectable partial R^2 as a function of sample size at the two bars used in the paper, with the largest observed partial R^2 and the discovery sample size marked.

7 What the design could have found

A null result is informative only in proportion to the smallest effect it could have detected. For a partial effect estimated on n observations, $t^2 \approx n R^2 / (1 - R^2)$, so at a fixed bar the smallest detectable partial R^2 falls as $1/n$. With 327 discovery weeks and the Bonferroni bar of $|t| > 2.394$, the floor is 1.72% of residual variance (Figure 4). The largest partial R^2 observed anywhere in the paper, across eight signals, two horizons, three mechanisms and two samples, is 0.009.

The scope of the result follows from this. It excludes effects large enough to be traded and is silent on smaller ones. A signal explaining a third of a percent of residual variance would be physically real and commercially useless, and this design could not distinguish it from zero. Two facts about scale make the null less surprising than the intuition that motivated the study. The EU-bound stock of gas at sea averages 3.69 million m^3 of LNG, which is about 2.21 bcm of gas or 2.3 days of EU demand at roughly 350 bcm per year. A one-standard-deviation change in that stock is 1.37 days of demand. The weekly spread has a standard deviation of \$11.22 per MMBtu. The question was therefore whether about a day and a half of marginal supply moves a price that responds to revisions in the weather forecast.

8 Discussion

The null is the outcome an efficient market predicts, and it is important to be precise about which question was asked. LNG cargo tracking is a commercial industry. Several vendors sell satellite-augmented AIS with proprietary destination inference to gas trading desks in real time. The signals in this paper are built from public terrestrial AIS and a free voyage archive, which is a coarser version of what the marginal trader already has. The test was therefore not whether the physical signal is informative, but whether public physical data is mispriced in a liquid benchmark at one- to four-week horizons. A negative answer to that question is close to what one would expect in advance. What the paper adds is evidence that the negative result is not an artefact of a defective pipeline, an unlucky specification, or a search that stopped at the first $t > 2$. The pipeline is validated against a published total, the specification was fixed before the fit, and the search was corrected for multiplicity and held out.

Three results stand independently of the null. First, the physical nowcasts converge on the naive mean from above, and for the one-week loadings target this follows from theory rather than from the particular models tried. The argument applies to any slow industrial flow series and should temper the expectation that a richer model will beat a moving average on one. Second, the two construction hazards of Section 3 apply to voyage-derived features in general, and they are easy to miss even when dates appear to have been handled correctly. The vintage log that detects them is inexpensive to maintain and, to our knowledge, rarely kept. Third, the outage rule of Section 5 is a usable monitor with a stated recall.

Several changes could alter the answer. A private information advantage, such as satellite coverage of the open ocean or a destination model that resolves the three-quarters of the stock this pipeline cannot, would change the question rather than the answer and would make the study a test of the vendor’s product. Within public data, the remaining untested approaches are joint rather than marginal fits (spike-and-slab or partial least squares over the collinear block), a fitted regime model in place of an observable split, distributed lags of the signals, and the twenty-six signal keys the panel does not use. Each has a weaker prior than the three mechanisms already tested, and each faces a larger multiplicity problem against a residual that has never exceeded 0.009 of variance. Any of them should be accompanied by its own pre-registration, its own multiplicity correction, and the untouched-holdout standard used here.

9 Limitations

No European anchorage history. The GFW feed carries port visits but no anchorage events, so European queue and absorption signals exist only from the 2026 live feed. Four of the five composite signals originally pre-registered as features for Section 6 turned out to have between 9 and 46 daily observations and were excluded by a documented amendment before any spread fit. The at-sea and US-side mechanism was tested on the decade; the EU-congestion mechanism could not be tested.

Destination resolution. Between 75% and 83% of the at-sea stock has no resolved destination. The EU-bound series is a minority observed slice whose definition depends on coverage. The unresolved band is carried as its own feature, and the ratio of the two was tested as H2. Attenuation from this measurement error biases coefficients toward zero and cannot be excluded as a contributing factor.

The capture gradient. NOAA-only capture of US export cargoes is 46% in 2020 and 74% in 2018 (Table 2). Early-year levels of every US signal are biased low. The walk-forward designs operate on within-window changes and the FWL scan on residuals, but a level-dependent non-linearity in the early years would be misread.

The coverage seam. The backfill feeds end on 2025-12-31 and 2026-02-21. The full-panel results in Appendix B include the thin live tail. In that sample 24% of the holdout weeks lie after the seam, and one full-panel finding, a change of sign in the H2 holdout coefficient, did not survive truncation and is not claimed. The primary sample is truncated at the seam.

Regime pooling. Signals are read from the pooled regime tag, which is the only tag spanning the decade. A small residual over-count where NOAA and GFW both observe the same US departure is documented in the project’s signal catalogue. A robustness run on the NOAA tag alone for the US-side signals was not performed.

The pre-registration trail. As Section 4 states, the git history does not independently establish that the log entries preceded the fits they govern.

Sample size and a dominant regime. The scored spread sample is 379 weeks, heavily autocorrelated, and 2022 alone spans a range larger than the remainder of the decade combined. The year-consistency condition and the holdout are the defences against this; a longer sample without a coverage seam would be preferable to either.

Vintage residual. 68 rows in the live tail changed after a next-day print (Section 3). They lie outside the modelling sample, but they show that the knowable bound is exact on event time and only approximately exact on observation time in a thin-coverage regime.

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A Signal definitions and pre-registered signs

This appendix lists the eight tanker signals used in Section 6, the aggregation applied when collapsing per-terminal or per-zone bands to one daily series, and the sign fixed before any fit. The sign convention is as follows. The spread $HH - TTF$ is negative in nearly every period, so a narrowing of the spread means a rise toward zero, which is a positive change.

signal	aggregation	mechanism	prior
gas in transit, EU-bound	sum over EU zones	more cargo landing in Europe soon; TTF falls	+
gas in transit, unresolved	sum	destination unknown; no directional prior	none
US loading rate	sum over US berths	gas leaves the US balance (HH firms) and reaches Europe (TTF falls)	+
EU discharge rate	sum over EU berths	Europe absorbing gas now; TTF falls	+
ballast returning to US	sum	tonnage returning to load; future exports rise	+
laden voyage age	n - weighted mean	cargo floating instead of landing; Europe stays tight	−
US load-queue wait	n - weighted mean	US cannot export; HH softens and Europe is short	−
net export pressure	n - weighted mean	$z(\text{loadings}) - z(\text{load queue})$; both components act in the same direction	+

Volumes are in m^3 of LNG. Loading and discharge rates are amortised m^3 per day across berth hours, so that a closed visit integrates to one cargo. Ages and waits are in days and hours respectively. The project’s signal catalogue documents the remaining twenty-six keys, the two bases, and the confidence columns (dispersion, open fraction and estimated fraction) that accompany every value.

A recorded inconsistency. The catalogue’s own text assigned net export pressure two incompatible signs in two places. The positive sign was adopted on the economic reasoning above and recorded as the prior. The data returned a negative point estimate at both horizons with $|t| \leq 1.6$, which neither confirms nor falsifies it.

B Full-panel results

The primary sample is truncated at the backfill horizon (2025-12-31). The same scans run to the data maximum (395 weeks) are reported here for completeness. They were the runs first recorded in the decision log, and they change no conclusion.

C Reproducibility, code, and the use of AI

Reproducibility. Every number in this paper is generated. The replays write machine-readable results (`make paper-results`). A generator converts them into the LaTeX tables and into a file of macros from which every in-prose number is drawn (`make paper-tables`).

Table 10: Model ladder, full panel.

h (wk)	model	n	MAE	RMSE	80% cov.	skill vs M0
1	M0	288	1.997	4.371	82.3%	–
1	M1	288	2.023	4.473	82.3%	–1.3%
1	M2	288	2.205	4.542	83.0%	–10.4%
4	M0	282	3.947	7.090	73.8%	–
4	M1	282	4.168	7.444	72.0%	–5.6%
4	M2	282	4.705	7.378	69.1%	–19.2%

Table 11: FWL partial effects, full panel. The largest $|t|$ here (1.72) belongs to the unresolved-destination band and falls to +0.24 on the primary sample. It was produced by the contaminated tail.

signal	prior	$h = 1$		$h = 4$		year cons.
		HAC t	partial R^2	HAC t	partial R^2	
gas in transit, EU-bound	+	+0.08	0.000	+1.03	0.004	75%
gas in transit, unresolved	none	-0.74	0.001	-1.72	0.009	62%
US loading rate	+	-0.19	0.000	+0.63	0.001	62%
EU discharge rate	+	+0.84	0.001	+0.46	0.000	62%
ballast returning to US	+	+0.14	0.000	+0.31	0.000	62%
laden voyage age	–	-0.04	0.000	-1.26	0.005	62%
US load-queue wait	–	+1.01	0.001	+0.55	0.001	38%
net export pressure	+	-0.82	0.001	-1.60	0.009	75%

The figures are drawn from the same tables the models read (**make paper-figures**). A number that is not generated cannot appear in the text, because the corresponding macro is undefined and the document does not compile. The underlying replays are **make a1-replay**, **a2-replay**, **a4-replay**, **a5-replay**, **a1-h1**, **b0-coverage** and **mechanisms-coverage**, together with their full-panel variants. The panel itself is produced by **make signals** followed by **make model-panel**. The source, the decision log and the test suite (591 tests at the time of writing) are available in the project repository.

Data and licences. AIS positions are from NOAA’s Marine Cadastre and Global Fishing Watch under their respective terms. EIA, FRED, GIE AGSI+ and Open-Meteo are free public APIs. The World Bank monthly series is licensed CC BY 4.0 and is the only price series redistributed with the code. The daily TTF series is obtained from a public quote feed under personal-use terms and is not redistributed; the loader retrieves it.

Use of AI. The author used an AI coding assistant (Anthropic’s Claude, through Claude Code) extensively in this project. A substantial fraction of the pipeline, the modelling harnesses, the tests and the first drafts of this text were written with its assistance. Every design decision, hypothesis, sign and acceptance bar in the decision log is the author’s. Every result was reproduced by the author from the stated **make** targets, and the author takes full responsibility for the content. The assistant was also used to audit the project’s own claims before this paper was written, which produced the corrections recorded in the final entries of the decision log.

Table 12: Mechanism tests, full panel. The holdout coefficient for H2 changes sign relative to the primary sample. In this sample 24% of the holdout weeks lie after the coverage seam, and the change of sign is not claimed as a finding.

hypothesis	prior	$\hat{\beta}$ (discovery)	HAC t	year cons.	partial R^2	$\hat{\beta}$ (holdout)
H2: EU share of at-sea stock	+	-2.64×10^0	-0.89	71%	0.003	8.23×10^0
H3: transit $\times$ EU storage	-	4.78×10^{-9}	+0.46	71%	0.001	6.73×10^{-10}
H4: transit, tight regime	+	-1.13×10^{-7}	-0.17	100%	0.000	9.21×10^{-7}
H4: transit, loose regime	+	-2.25×10^{-7}	-1.00	100%	0.009	-

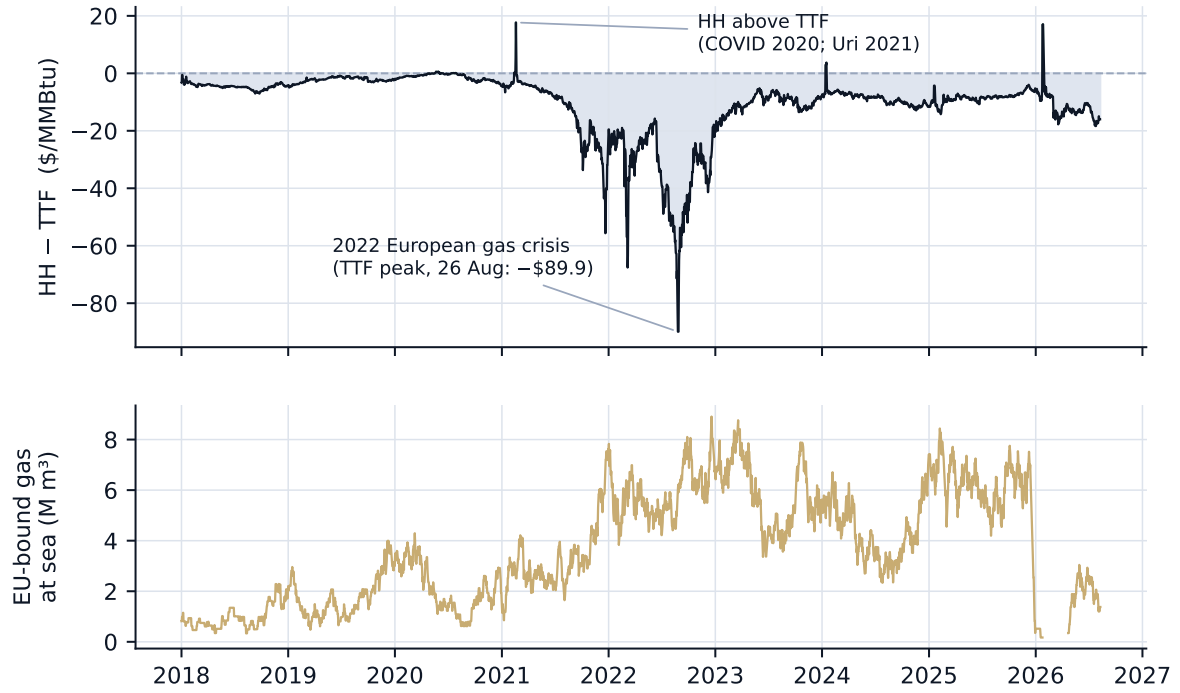


Figure 5: Figure 1 extended to the data maximum. The 91% step in the at-sea stock at the start of 2026 is the coverage seam and not a physical event.

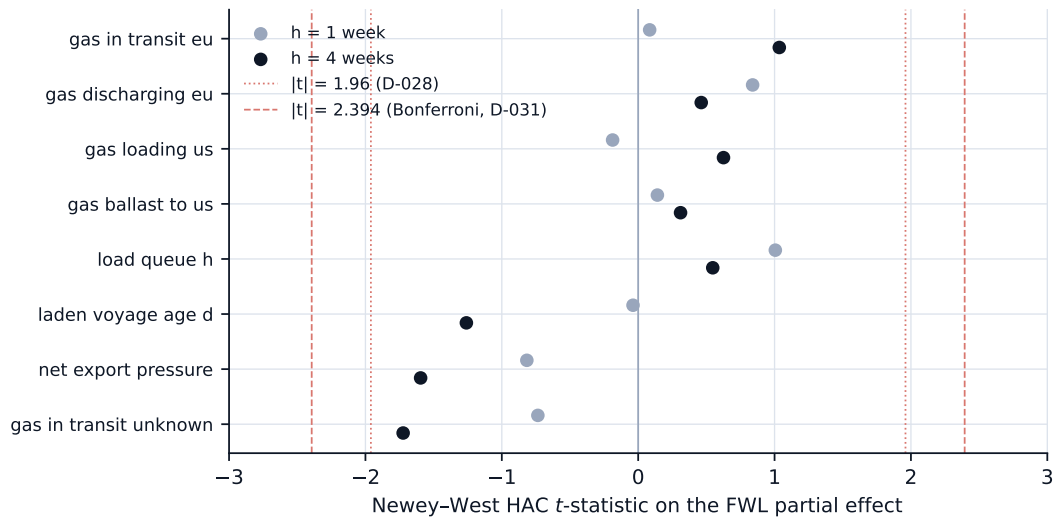


Figure 6: Figure 3 on the full panel.